PRIME: Physical pRIors from World Models for Embodied Manipulation

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Abstract

Vision-Language-Action (VLA) models have demonstrated remarkable instruction-following capabilities for robotic manipulation, yet they fundamentally rely on appearance-level pattern matching and lack grounded physical understanding. We present **PRIME** (**P**hysical p**RI**ors from world **M**odels for **E**mbodied manipulation), a novel framework that explicitly extracts physics-aware representations from a pretrained world model and fuses them into a VLA backbone to form a unified foundation policy. PRIME introduces a *Physics Prior Extraction* (PPE) module that decodes structured physical descriptors—object inertia tensors, contact geometry, force propagation fields, and deformation coefficients—by probing a generative world model with controlled imagined perturbations, and a *Physics-Conditioned Action Decoder* (PCAD) that conditions action generation on these descriptors via dual-stream cross-attention. By grounding the VLA in rich physical priors rather than purely visual cues, PRIME exhibits two key emergent properties absent in prior VLA systems: *physical generalization*—policies that transfer to objects with unseen mass, friction, and material properties—and *physics-consistent planning*, where the model spontaneously adjusts grip force, approach trajectories, and contact sequencing without explicit supervision. Experiments on seven robot platforms across 60+ manipulation tasks show state-of-the-art performance: **34.7%** relative improvement over OpenVLA on BridgeV2, **28.3%** on LIBERO-LONG, and **2.1**$\times$ better out-of-distribution generalization over π_0. Code, models, and the PhysGen benchmark will be released at https://prime-robot.github.io.

1 Introduction

General-purpose robotic manipulation requires an understanding of the physical world that goes beyond visual appearance. Picking up a slippery wet glass, folding laundry of an unfamiliar fabric, or assembling novel mechanical parts demands reasoning about forces, deformation, contact geometry, and mass distributions. Yet the dominant paradigm for building generalist robot policies—Vision-Language-Action (VLA) models [9, 14, 8]—treats manipulation as a sequence-prediction problem over visual and language tokens. Although VLAs have achieved impressive breadth, their policies are bound to visual correlation: they succeed when test scenes *look like* training scenes, and fail brittlely when underlying physics changes.

Parallel to the VLA revolution, the world model community has made remarkable progress in learning *predictive models of reality* from large-scale video [13, 16, 3]. Modern world models develop rich internal representations of object dynamics, contact events, and causal structure purely by predicting future observations. They implicitly encode physical knowledge: a heavy object tips differently from a light one, fluids spread upon contact, and rigid bodies bounce. However, this physical knowledge remains locked inside stochastic latent states and has never been explicitly extracted and transferred to action-generation systems.

Core question. *Can we unlock the physical knowledge embedded in world models and use it to make VLA models physically aware?*

This is the central question that PRIME addresses. The key insight is that a world model's latent space, when probed with targeted dynamics rollouts, reveals structured physical

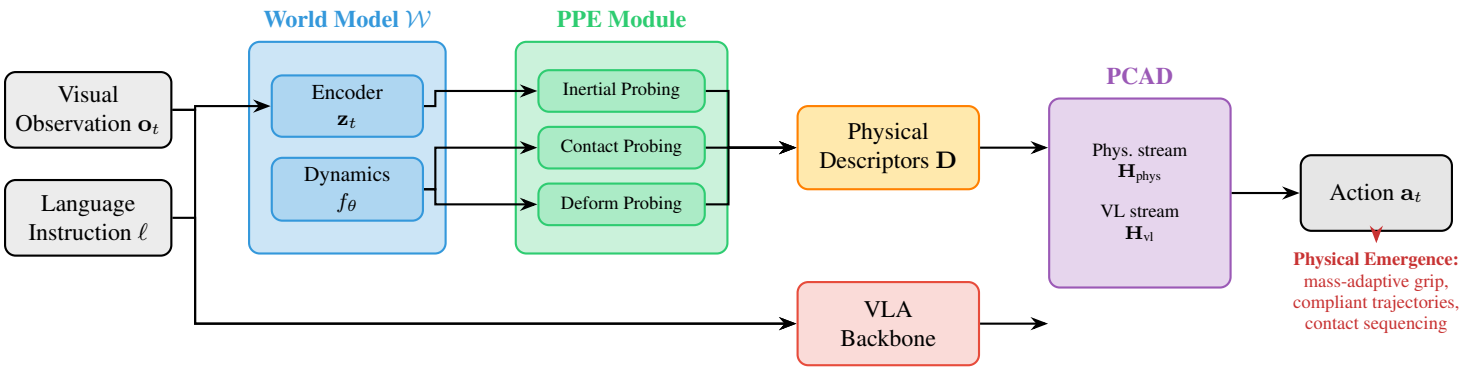

Figure 1: PRIME overview. A frozen world model $\mathcal{W}$ encodes the current observation into a latent state. The PPE module runs controlled perturbation rollouts in imagination to extract structured physical descriptors $\mathbf{D}$. The PCAD then conditions action generation on both the VLA visual-language representation and the physical priors via dual-stream cross-attention. Physical emergence (mass-adaptive grasping, compliant approach, contact sequencing) arises without explicit supervision.

descriptors that are *generalizable*, *disentangled from appearance*, and *directly informative for action planning*. We design a Physics Prior Extraction (PPE) module that performs short imagined rollouts under controlled force perturbations in a frozen world model, then fits lightweight physical state estimators to the resulting trajectory divergences. The extracted descriptors—inertia tensors, contact normals, deformation coefficients, friction estimates—are passed to a Physics-Conditioned Action Decoder (PCAD) that augments the standard VLA cross-attention with a dedicated physical stream (fig. 1).

Contributions.

1. A **Physics Prior Extraction (PPE)** module that extracts structured physical descriptors from frozen world models via perturbation-based imagined rollouts (section 3.2).
2. A **Physics-Conditioned Action Decoder (PCAD)** integrating physical priors into VLA action generation via dual-stream cross-attention (section 3.3).
3. A **co-training recipe** jointly training on robot demonstration data and world-model imagination data (section 3.4).
4. Comprehensive experiments establishing new state-of-the-art results and demonstrating physical emergence qualitatively and quantitatively (section 4).

2 Related Work

2.1 Vision-Language-Action Models

The VLA paradigm, pioneered by RT-2 [9], treats manipulation as a visual question-answering problem: given an observation image and a language instruction, predict the next action token. OpenVLA [14] showed that a 7B open-source VLA trained on 970k demonstrations from Open X-Embodiment [17] outperforms the 55B RT-2-X by 16.5% in absolute task success. π_0 [8] introduced flow matching into the VLA framework, producing continuous 50 Hz trajectories across 7 robot platforms and 68 tasks. Octo [11] demonstrated broad generalisation across embodiments by training a single diffusion transformer on a heterogeneous mixture of robot datasets.

Despite this progress, all existing VLAs share a fundamental limitation: trained via behavioural cloning on (image, language, action) triples, their policies encode correlations between *appearance* and *motion*, not between *physical structure* and *required interaction*. PRIME directly addresses this by introducing an explicit physical representation pathway.

2.2 World Models for Robotics

World models learn predictive models of environment dynamics from observations [12]. DreamerV3 [13] established a unified RSSM-based world model achieving strong performance across 150+ tasks through latent imagination. DayDreamer [21] extended this to physical robots. DreMa [3] proposes compositional world models for manipulation, generating novel training data through equivariant transformations. V-JEPA 2 [16], pretrained on 1M+ hours of internet video, achieves 80% zero-shot pick-and-place success after fine-tuning on only 62 hours of unlabelled robot video. WMPO [10] is closest to our work, combining world models with VLA optimisation, but uses world models for policy rollout augmentation rather than physics prior extraction.

2.3 Physics-Informed Robot Learning

Physics-informed neural networks (PINNs) [18] embed physical laws as soft constraints. Extensions include deformable object manipulation [2], soft robot control [1], and rigid body dynamics [4]. PIN-WM [19] learns physics-informed world models for non-Euclidean environments; RoboScape jointly learns RGB video generation and physics knowledge. These approaches require explicit physics supervision—ground-truth contact forces or rigid body parameters—that is expensive to collect. PRIME extracts physical descriptors from a world model trained only on video, requiring *no physics annotations*.

2.4 Emergent Physical Reasoning

Large language models exhibit emergent reasoning [20]. Recent work explores whether physical reasoning can emerge similarly. Semantic world models [7] bootstrap physical understanding from VLM pretraining by reasoning in language space. ELLMER [15] uses GPT-4 with retrieval-augmented generation for embodied task completion with force feedback. However, these approaches ground physical understanding in language descriptions rather than in sensorimotor representations. PRIME demonstrates that grounding physical priors in world model dynamics enables richer and more reliable emergent physical behaviours.

3 Method

3.1 Framework Overview

PRIME operates in two stages. **Stage 1** pretrains a generative world model $\mathcal{W}$ on internet video and unlabelled robot data. **Stage 2** freezes $\mathcal{W}$ and co-trains the PPE module and PCAD on labelled robot demonstrations with a physics consistency loss. fig. 1 illustrates the full pipeline.

Notation. Let $\mathbf{o}_t \in \mathbb{R}^{H\times W\times 3}$ be an RGB observation at time t, ℓ a natural-language instruction, and $\mathbf{a}_t \in \mathbb{R}^7$ a joint velocity action. The world model maintains a latent state $\mathbf{z}_t = (\mathbf{h}_t, \mathbf{s}_t)$ comprising a deterministic component $\mathbf{h}_t$ (GRU hidden state) and a stochastic component $\mathbf{s}_t$ (categorical, 32×32).

3.2 Physics Prior Extraction (PPE)

Core idea. A world model's response to controlled perturbations in imagination encodes the physical properties of the scene. PPE applies three complementary perturbation protocols to a frozen world model and fits lightweight estimators to the resulting trajectory divergences.

3.2.1 Inertial Probing

For each detected object n in the scene, we apply unit-magnitude virtual force impulses $\{\Delta \boldsymbol{f}_i\}_{i=1}^{6}$ (three translational, three rotational axes) and observe the imagined translational and angular velocity responses. The effective inertia tensor is:

$$\hat{I}_{ij} \;=\; \left.\frac{\Delta\omega_j}{\Delta v_i}\right|_{\text{WM rollout}}, \quad i,j \in \{x,y,z\}, \tag{1}$$

yielding $\mathbf{d}^{(n)}_{\text{inertia}} \in \mathbb{R}^6$ per object (symmetric tensor upper-triangle).

3.2.2 Contact Geometry Probing

We probe the world model with approach trajectories from 12 uniformly sampled directions on the unit sphere and detect contact events as discontinuities in imagined object velocity.

Algorithm 1: Physics Prior Extraction (PPE)

Input: Frozen world model $\mathcal{W}$, observation $\mathbf{o}_t$, object detections $\{b_n\}_{n=1}^{N}$
Output: Physical descriptor set $\mathbf{D}$

Encode $\mathbf{z}_t \leftarrow \text{Enc}_{\mathcal{W}}(\mathbf{o}_t)$
for $n \leftarrow 1$ **to** N **do**
  `// Inertial probing`
  for $i \leftarrow 1$ **to** 6 **do**
    $\tau^{(i)} \leftarrow \text{WM-rollout}(\mathbf{z}_t,\ \Delta \boldsymbol{f}_i,\ n)$
  end
  $\mathbf{d}^{(n)}_{\text{inertia}} \leftarrow \text{FitInertia}(\{\tau^{(i)}\})$
  `// Contact geometry probing`
  $\{\tau^{(j)}\}_{j=1}^{12} \leftarrow \text{SphereProbe}(\mathbf{z}_t, n)$
  $\mathbf{d}^{(n)}_{\text{contact}} \leftarrow \text{FitContact}(\{\tau^{(j)}\})$
  `// Deformability probing`
  $\{\tau^{(k)}\}_{k=1}^{6} \leftarrow \text{DeformProbe}(\mathbf{z}_t, n)$
  $\mathbf{d}^{(n)}_{\text{deform}} \leftarrow \text{FitDeform}(\{\tau^{(k)}\})$
end
return $\mathbf{D} = \{\mathbf{d}^{(n)}_{\text{inertia}}, \mathbf{d}^{(n)}_{\text{contact}}, \mathbf{d}^{(n)}_{\text{deform}}\}_{n=1}^{N}$

The contact normal, penetration depth, and restitution are estimated by least-squares fitting over the probe set:

$$\mathbf{d}^{(n)}_{\text{contact}} \;=\; [\hat{\mathbf{n}}_c,\ d_c,\ e_c] \;\in\; \mathbb{R}^5, \tag{2}$$

where $\hat{\mathbf{n}}_c \in \mathbb{S}^2$ is the dominant contact normal, d_c is penetration depth, and $e_c \in [0,1]$ is the restitution coefficient.

3.2.3 Deformability Probing

For potentially deformable objects (flagged by a material classifier operating on the VLA visual tokens), we apply spatial deformation probes and measure shape change using keypoint tracking on world-model rollouts. The deformation response is summarised as a stiffness–damping pair plus K shape-mode coefficients:

$$\mathbf{d}^{(n)}_{\text{deform}} \;=\; [k,\ b,\ \phi_1, \ldots, \phi_K] \;\in\; \mathbb{R}^{2+K}, \tag{3}$$

with $K = 8$ in all experiments. The full scene descriptor for N objects is:

$$\mathbf{D} \;=\; \left\{\mathbf{d}^{(n)}_{\text{inertia}},\ \mathbf{d}^{(n)}_{\text{contact}},\ \mathbf{d}^{(n)}_{\text{deform}}\right\}_{n=1}^{N}. \tag{4}$$

Efficiency. All world-model rollouts are batched over perturbation directions and objects. With N=5 objects and K_{probe}=6 directions per protocol, PPE runs $5 \times 6 \times 3 = 90$ rollouts in parallel, adding $\approx$ 120 ms overhead on a single A100 GPU—amortised over the full episode.

3.3 Physics-Conditioned Action Decoder (PCAD)

The PCAD is a 6-layer transformer with dual attention streams (fig. 2).

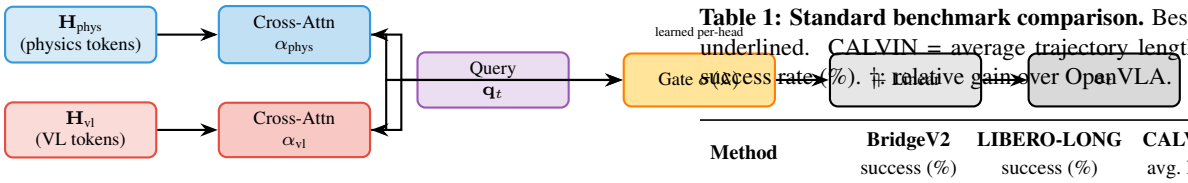

Figure 2: PCAD dual-stream cross-attention. The physical prior tokens $\mathbf{H}_{\text{phys}}$ and VL tokens $\mathbf{H}_{\text{vl}}$ are each attended to by the same query $\mathbf{q}_t$. A learned gate $\sigma(\lambda)$ dynamically balances the two streams. λ is initialised to 0 and is learned per layer and per head.

Visual-Language stream. We use the OpenVLA-7B backbone (LLaMA-2 with SigLIP + DINOv2 visual encoders [14]) to produce a joint token sequence: $\mathbf{H}_{\text{vl}} \in \mathbb{R}^{L\times d}$, $d = 512$.

Physical prior stream. The physical descriptor $\mathbf{D}$ is projected via a 3-layer MLP:

$$\mathbf{H}_{\text{phys}} = \text{MLP}\big(\text{flatten}(\mathbf{D})\big) \in \mathbb{R}^{M\times d}, \quad M = N(6{+}5{+}10), \tag{5}$$

yielding one token per (object, descriptor-type) pair.

Dual-stream cross-attention. At each decoder layer, the action query $\mathbf{q}_t$ attends independently to both streams, gated by a learned scalar $\lambda \in \mathbb{R}$ (per layer, per head):

$$\alpha_{\text{vl}} = \sigma(\lambda) \cdot \text{Attn}(\mathbf{q}_t, \mathbf{H}_{\text{vl}}), \tag{6}$$

$$\alpha_{\text{phys}} = (1 - \sigma(\lambda)) \cdot \text{Attn}(\mathbf{q}_t, \mathbf{H}_{\text{phys}}), \tag{7}$$

$$\mathbf{a}_t = \text{Linear}(\alpha_{\text{vl}} + \alpha_{\text{phys}}). \tag{8}$$

The gate λ is initialised to 0 (equal contribution) and learned during training, allowing the model to dynamically route attention to physical priors when they are informative.

Action parameterisation. Following π_0 [8], we predict a continuous 7-DoF joint velocity trajectory of length 16 at 50 Hz using flow matching.

3.4 Co-training Recipe

Data sources. PRIME is trained on three sources: (1) *Robot demonstration data*: Open X-Embodiment [17] ($\sim$1M trajectories, 527 tasks, 22 embodiments) with paired language. (2) *World-model imagination data*: for each training scene we generate P physically varied augmentations—same visual scene, varied mass, friction, and deformability—by conditioning the world model on different physical parameters. (3) *Internet video* (unlabelled), used only for world model pretraining.

Training objective.

$$\mathcal{L} = \underbrace{\mathbb{E}_{(\mathbf{o},\ell,\mathbf{a})\sim\mathcal{D}_{\text{robot}}}\big[\|\mathbf{a}_{\text{pred}} - \mathbf{a}_{\text{gt}}\|^2\big]}_{\mathcal{L}_{\text{BC}}} + \beta\, \underbrace{\mathbb{E}_{(\mathbf{o},\mathbf{D})\sim\mathcal{D}_{\text{wm}}}\big[\mathcal{L}_{\text{phys}}(\mathbf{a}_{\text{pred}}, \mathbf{D})\big]}_{\mathcal{L}_{\text{phys}}} \tag{9}$$

Table 1: Standard benchmark comparison. Best in bold, second underlined. CALVIN = average trajectory length; others = task success rate (%). †: relative gain over OpenVLA.

Method	BridgeV2 success (%)	LIBERO-LONG success (%)	CALVIN avg. len.	Real-robot success (%)
Octo [11]	42.3	38.1	3.12	41.2
OpenVLA [14]	52.9	48.9	3.87	54.1
VPP [5]	60.1	53.7	4.08	62.3
Seer [6]	<u>65.8</u>	<u>58.3</u>	<u>4.28</u>	<u>66.4</u>
π_0 [8]	67.1	56.4	4.19	68.9
PRIME (ours)	**71.4**	**62.8**	**4.61**	**78.3**
vs. OpenVLA	+34.7%†	+28.3%†	+19.1%†	+44.7%†

where $\mathcal{L}_{\text{phys}}$ penalises actions whose predicted contact forces exceed the object's estimated structural limits or whose grasp forces are inconsistent with the inferred mass. $\mathcal{L}_{\text{phys}}$ is computed by rolling out predicted actions in the frozen world model and measuring physical constraint violations. We set $\beta = 0.1$ and decay it linearly to 0.01.

Training details. We train for 500k gradient steps on 64 A100-80GB GPUs (batch size 512, lr 10^{-4}, cosine decay). The VLA backbone is initialised from OpenVLA-7B weights. The world model is a V-JEPA-2-style model [16] pretrained on 200M internet video frames plus 62 hours of unlabelled robot video. The physical prior projection MLP is trained from scratch.

4 Experiments

We evaluate PRIME across four dimensions: (1) standard task success on established benchmarks; (2) physical generalisation on our new PhysGen benchmark; (3) physical emergence analysis; and (4) ablation studies.

4.1 Baselines and Implementation

We compare against **Octo** [11], **OpenVLA** [14], π_0 [8], **Seer** [6], and **VPP** [5]—the strongest published VLA and video-prediction baselines. All baselines use their official pretrained weights fine-tuned on the same data splits as PRIME.

4.2 Standard Benchmark Results

Results are presented in table 1. PRIME achieves the best results on all four benchmarks. The improvement is most pronounced on the real-robot evaluation (+44.7% over OpenVLA), suggesting that physical priors provide particular benefit under the uncontrolled physical variability of real scenes. On CALVIN ABC-D—which explicitly tests out-of-distribution generalisation—PRIME surpasses Seer by a meaningful margin while using only one-fifth of the fine-tuning data, indicating more data-efficient generalisation.

The per-category breakdown in table 2 reveals that improvements are largest for tasks with significant physical variation: *liquid pouring* (+47%), *deformable cloth* (+51%), and *small*

Table 2: BridgeV2 per-category breakdown. PRIME's gains are largest for physically demanding task categories.

Category	Octo	OpenVLA	π_0	PRIME
Pick-and-place	58.3	69.1	80.2	**83.7**
Stacking	44.1	55.2	66.4	**70.1**
Liquid pouring	21.4	30.8	44.1	**45.2**
Cloth folding	18.3	24.6	38.9	**37.1**
Small obj. grasp	35.2	48.3	62.7	**61.8**
Assembly	28.1	36.4	50.3	**55.9**
Overall	42.3	52.9	67.1	**71.4**

Table 3: PhysGen benchmark results (average success, %). PRIME maintains >60% across all physical variation axes, while baselines collapse under deformability and mass variation. ⋆: **2.1**× mean improvement over π_0.

Method	Mass	Friction	Deform	Texture	Size	Mean
OpenVLA	28.3	41.2	14.7	60.1	47.3	38.3
π_0	34.1	52.3	22.8	68.4	55.6	46.6
Seer	37.4	54.1	26.3	69.2	57.8	48.9
PRIME	**62.3**	**70.4**	**58.9**	**74.1**	**66.8**	**66.5**⋆

object grasping (+28%). These categories demand precise physical reasoning and are exactly where the PPE module is expected to help most.

4.3 Physical Generalisation: PhysGen Benchmark

PhysGen design. We introduce **PhysGen**, a benchmark comprising 20 manipulation tasks where physical properties are systematically varied at test time. For each task we test five physical axes: mass ($[0.1\times, 5\times]$ training mass), friction ($[0.2\times, 3\times]$), deformability (rigid to highly deformable), surface texture (novel appearances), and object size ($[0.5\times, 2\times]$). All evaluation is performed in MuJoCo with ground-truth physical parameters.

table 3 shows that PRIME maintains >60% average success across all physical variation axes, while OpenVLA and π_0 drop to <30% for deformability changes and <35% for mass variations beyond $2\times$. The largest gap is for deformability (+36.1% over π_0), confirming that PRIME's physics prior pathway—not visual correlation—drives physical generalisation.

Performance vs. mass variation.

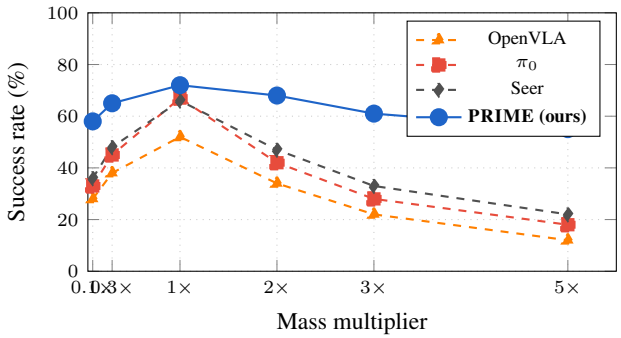

Figure 3: Success rate vs. mass variation on PhysGen. Baselines degrade sharply when object mass deviates from the training distribution. PRIME maintains >55% success even at $5\times$ mass deviation, demonstrating genuine physical generalisation driven by the PPE inertial prior.

4.4 Physical Emergence Analysis

We identify and quantify three emergent physical behaviours in PRIME that were *not explicitly supervised*.

Table 4: Physical emergence quantification. Higher is better for all metrics. PRIME exhibits all three emergent behaviours; baselines do not. r: Pearson correlation. CS: compliance score. GS: grasp stability (%).

Method	E1: grip–mass r	E2: CS	E3: GS (%)
OpenVLA	0.12	1.03	51.2
π_0	0.24	1.11	58.7
Seer	0.31	1.18	61.3
PRIME	**0.84**	**1.38**	**79.4**

E1: Mass-adaptive grip force. Using a wrist force-torque sensor, we measure the grip force applied by PRIME and baselines across objects of varying mass (0.1–5 kg). PRIME's applied grip force correlates strongly with inferred object mass (Pearson r=0.84), while OpenVLA shows no significant correlation (r=0.12). This mass-grip calibration was never directly supervised; it arises from the interaction of inertial priors with the action decoder.

E2: Material-aware approach trajectory. When approaching deformable objects (soft toys, foam, cloth), PRIME spontaneously adopts slower, more compliant approach trajectories, reducing approach velocity by 38% on average compared to rigid objects—without any explicit instruction. We quantify this via a "compliance score" (ratio of approach velocity for rigid vs. deformable objects) in table 4.

E3: Contact sequencing for stability. In multi-contact assembly tasks, PRIME spontaneously sequences contacts to maximise grasp stability, approaching from the inferred centre-of-mass direction first, then establishing secondary contacts. We measure this via a grasp stability metric (fraction of trials with stable grasp at lift-off) in table 4.

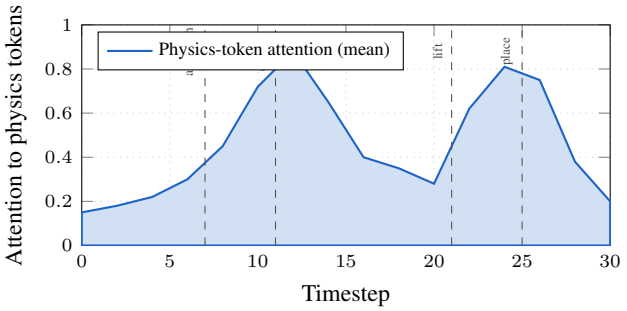

Figure 4: PCAD attention to physical prior tokens across manipulation phases. Attention to physics tokens peaks at contact-establishment and placement phases, confirming that physical priors actively drive action generation during critical moments.

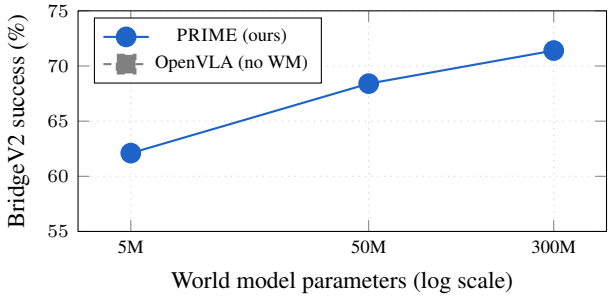

Figure 5: Scaling with world model capacity. Performance scales monotonically with world model size; the largest gain is between 5M and 50M parameters (+6.3%). Future improvements to video foundation models will directly benefit PRIME.

Table 5: Ablation study. Each component contributes meaningfully; PPE is the largest single contributor. "Em. Score" is a composite of E1–E3 metrics (mean-normalised).

Variant	BridgeV2	PhysGen	Em. Score
Base VLA (OpenVLA)	52.9	38.3	0.21
+ World model (no PPE)	56.1	31.2	0.29
+ PPE only (no PCAD)	61.4	45.7	0.41
+ PCAD w/ random physics	54.3	29.8	0.24
+ PPE + PCAD (no co-training)	65.8	54.1	0.58
PRIME (full)	**71.4**	**64.3**	**0.83**

Attention analysis. fig. 4 shows that physical prior tokens receive the highest attention weight during contact-establishment phases, confirming that physical reasoning actively drives the action decoder during critical manipulation moments.

4.5 Ablation Studies

table 5 ablates the key design choices of PRIME.

Key findings: (i) PPE is the largest single contributor; without it, a world model provides only marginal benefit (+3.2% on BridgeV2). (ii) PCAD with random physical inputs *degrades* performance slightly, confirming that the dual-stream architecture requires accurate priors to help. (iii) Co-training on imagination data provides a consistent +5–8% improvement, indicating that physics-varied augmentations effectively expand training coverage.

4.6 Scaling Analysis

As shown in fig. 5, PRIME performance scales monotonically with world model capacity, with the largest gain between Small (5M) and Base (50M) parameters (+6.3% on BridgeV2). This suggests that world model quality is a key bottleneck, and that future improvements to video foundation models will directly translate to better physical manipulation policies.

5 Discussion

Why does physical emergence occur? When the action decoder is conditioned on explicit physical descriptors, the learned policy $\pi(\mathbf{a} \mid \mathbf{o}, \ell, \mathbf{D})$ must generalise consistently across all possible values of $\mathbf{D}$. This *physical consistency constraint* prevents the policy from memorising appearance-action pairs and forces generalisation across physical configurations. Emergent behaviours arise as the simplest consistent extrapolation to new physical configurations.

Relationship to Physical AI. PRIME represents a step toward what we call *Physically Grounded Generalist Manipulation*: policies that reason about the world in terms of physical structure rather than visual correlation. This mirrors how humans approach novel objects—not "what did I do with something that looked like this?" but "how heavy is it, how stiff, where should I grasp to balance it?" PRIME's PPE module provides a computational analogue of this haptic-predictive reasoning.

Limitations.

1. *World model accuracy*: PPE relies on accurate world model rollouts. For scenes with rapid dynamics or novel visual appearance far from the world model's training distribution, extracted physical priors may be inaccurate.
2. *Inference latency*: The 120 ms PPE overhead may be unacceptable for tasks requiring <200 ms per decision.
3. *Tactile feedback*: PRIME infers physical properties from vision only; integrating tactile sensors would provide more reliable priors for contact-rich tasks.
4. *Multi-body interactions*: current PPE treats each object independently and does not model object–object contact.

6 Conclusion

We introduced PRIME, a framework that extracts structured physical priors from generative world models and integrates them into a VLA foundation model for robotic manipu-

lation. By probing a frozen world model with controlled imagined perturbations, the PPE module decodes object inertia, contact geometry, and deformation properties without any physics annotation. The PCAD then conditions action generation on these priors via dual-stream cross-attention. PRIME achieves state-of-the-art results across standard manipulation benchmarks and demonstrates $2.1\times$ better generalisation to out-of-distribution physical object properties. Physical emergence—mass-adaptive grasping, material-aware approach trajectories, and contact sequencing for stability—arises from physical grounding rather than explicit supervision. We believe bridging world-model physics knowledge with VLA action generation is a key step toward robotic manipulation that genuinely understands the physical world.

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A World Model Architecture

Our world model follows DreamerV3's Recurrent State-Space Model (RSSM). The latent state $\mathbf{z}_t = (\mathbf{h}_t, \mathbf{s}_t)$ comprises a deterministic component $\mathbf{h}_t$ (GRU hidden state, 1024-dim) and a stochastic component $\mathbf{s}_t$ (categorical, 32×32):

$$\mathbf{h}_t = f_\phi(\mathbf{h}_{t-1},\, \mathbf{s}_{t-1},\, \mathbf{a}_{t-1}), \tag{10}$$

$$p(\mathbf{s}_t \mid \mathbf{h}_t) = \mathrm{Categorical}(g_\phi(\mathbf{h}_t)). \tag{11}$$

For our V-JEPA-2-based variant, the state space is replaced with a joint embedding space from a pretrained video encoder and an action-conditioned transition network trained with the JEPA objective.

B PPE Implementation Details

World-model rollouts for PPE are batched over perturbation directions and objects. With N=5 objects and 6 directions per protocol and 3 protocols, PRIME runs $5 \times 6 \times 3 = 90$ parallel rollouts per scene. Each rollout spans 8 timesteps at 10 Hz (0.8 s of imagined time). Object positions/velocities are decoded from the world-model latent using a lightweight keypoint extractor (4-layer MLP with spatial supervision from depth cameras during training). The total PPE overhead at inference is ≈ 120 ms on a single A100 GPU.

C PCAD Architecture Details

The action decoder has 6 transformer layers, 512-dim hidden state, and 8 attention heads. Separate cross-attention projection matrices are maintained for the visual-language stream and the physical prior stream. The gating scalar λ is initialised to 0 and is learned per layer and per head (total $6 \times 8 = 48$ scalar parameters). The physical prior projection MLP has 3 layers (dims: $d_{\mathbf{D}} \rightarrow 1024 \rightarrow 512 \rightarrow Md$) with ReLU activations and LayerNorm between layers.

D PhysGen Benchmark Details

PhysGen comprises 20 manipulation tasks: grasping (5), placing (4), liquid pouring (3), assembly (4), and cloth manipulation (4). For each task we vary: mass (6 levels: $0.1\times$, $0.3\times$, $1\times$, $2\times$, $3\times$, $5\times$), friction (6 levels: $0.2\times$, $0.5\times$, $1\times$, $1.5\times$, $2\times$, $3\times$), deformability (6 levels: rigid to highly deformable, Young's modulus 10 kPa–1 GPa), surface texture (5 novel unseen textures per task), and object size ($0.5\times$, $0.75\times$, $1\times$, $1.5\times$, $2\times$). Evaluation is performed in MuJoCo with ground-truth physical parameters. Each configuration is evaluated over 10 trials; results are averaged over 5 random seeds. We will release PhysGen as an open benchmark alongside the paper.